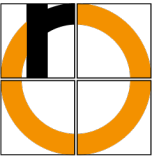


Computer Science Fundamentals

Graph Theory – Introduction

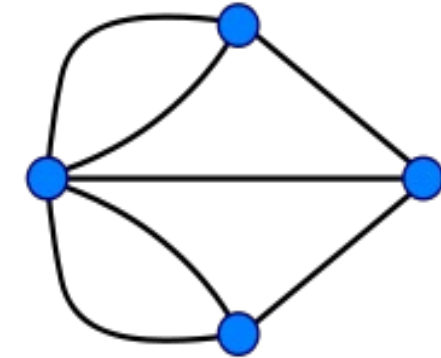
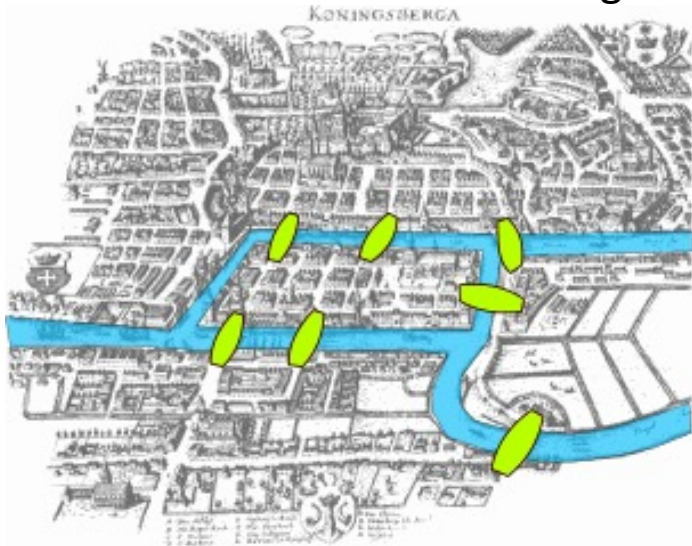
Technische Hochschule Rosenheim
Winter 2025/26
Prof. Dr. Jochen Schmidt

Seven Bridges of Königsberg (*Königsberger Brückenproblem*)

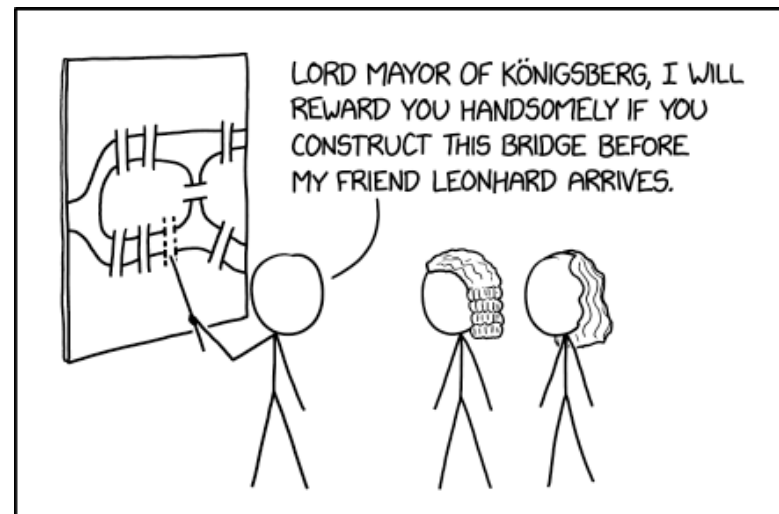


Leonhard Euler 1736:

Is there a circular route through Königsberg that crosses each of the seven bridges over the Pregel exactly once?

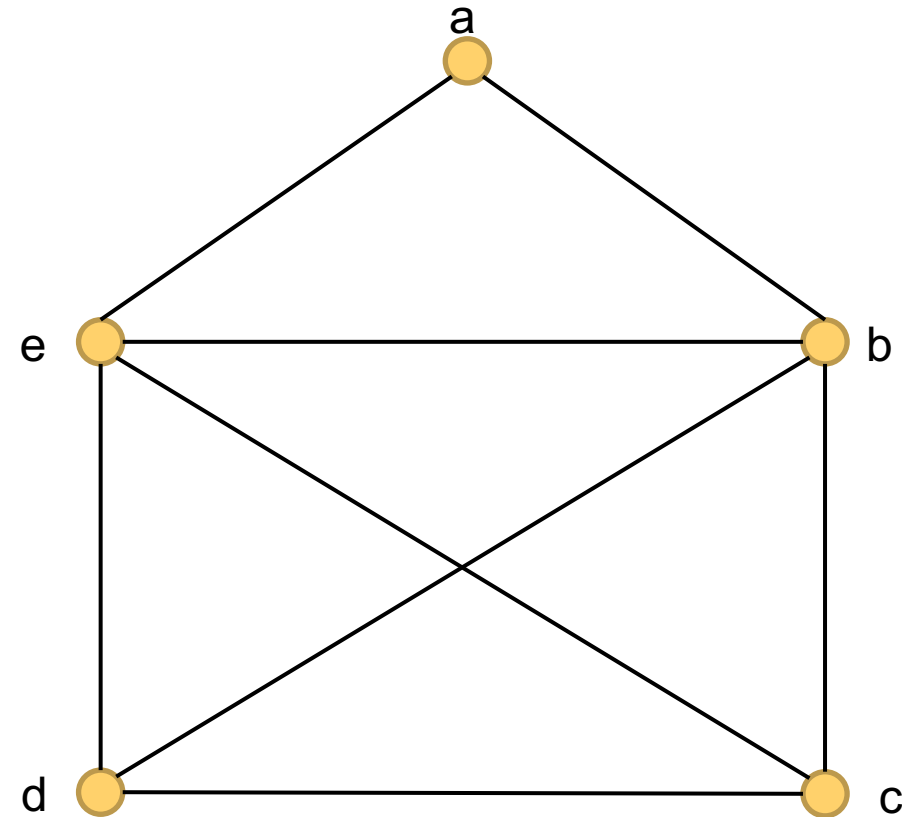
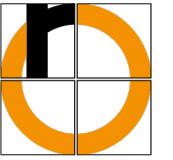


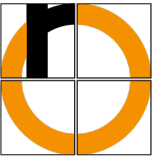
Is there a circular route that contains each edge exactly once?



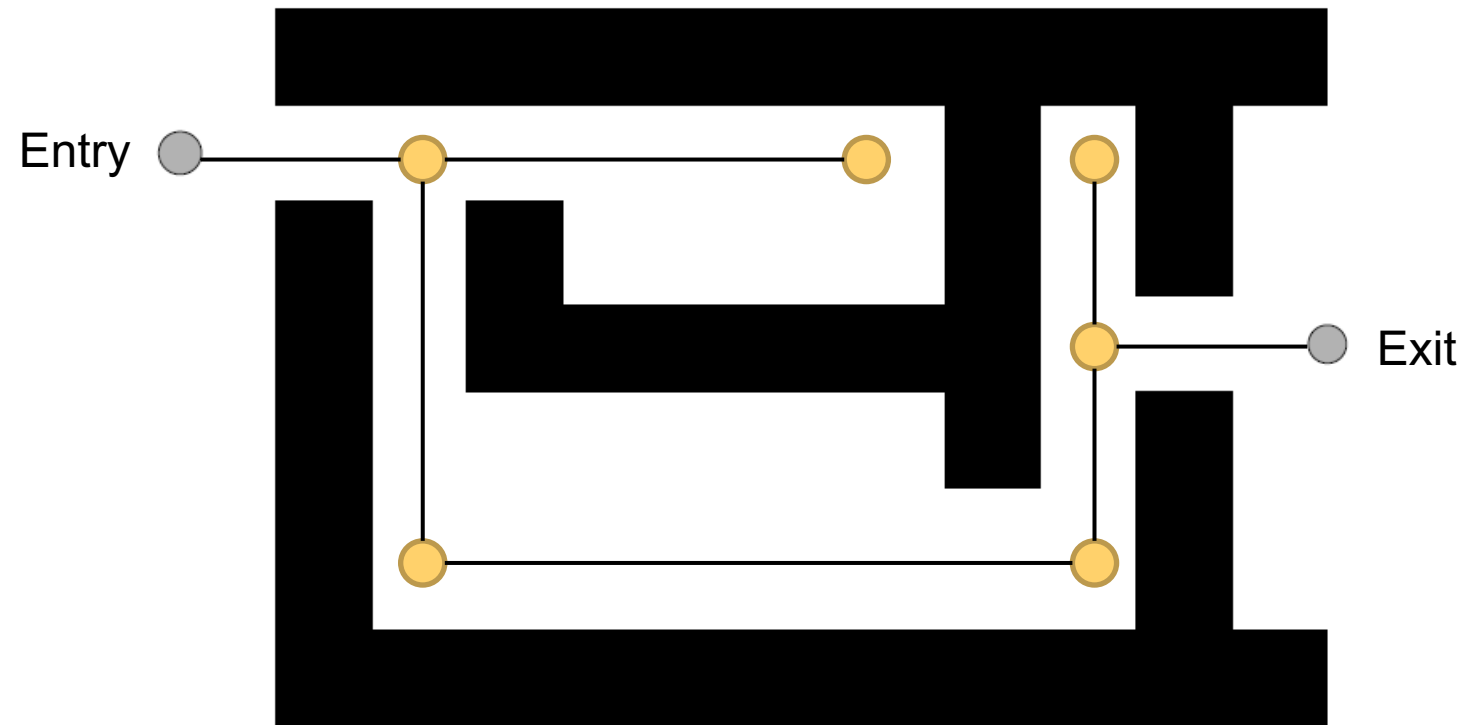
I TRIED TO USE A TIME MACHINE TO CHEAT ON MY ALGORITHMS
FINAL BY PREVENTING GRAPH THEORY FROM BEING INVENTED.

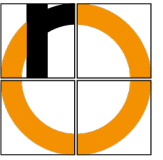
<https://xkcd.com/2694>, Königsberg, CC BY-NC 2.5





Find a way through the labyrinth!





Vertices (*Knoten*): $V = \{a, b, c, d\}$

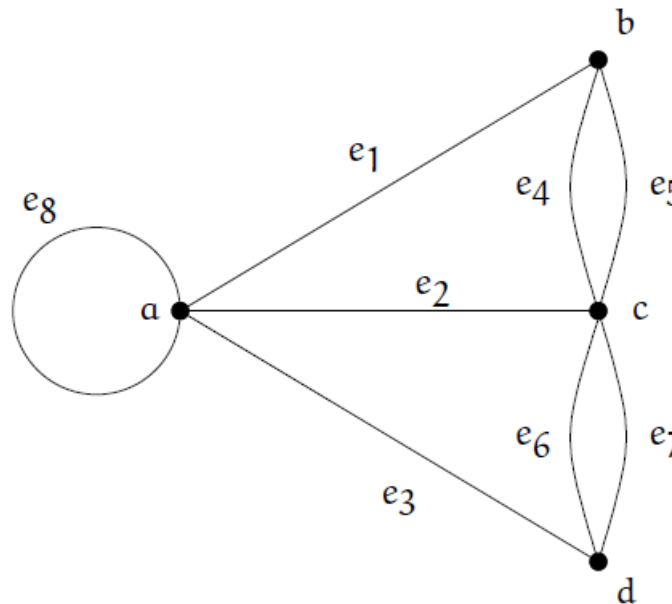
Edges (*Kanten*): $E = \{e_1, e_2, \dots, e_8\}$

Incidence mapping (*Inzidenzabbildung*):

$I = \{(e_1, \{a, b\}), (e_2, \{a, c\}), \dots, (e_8, \{a\})\}$
defines which edges connect which vertices

Loop (*Schlinge*):

Edge is incident on a single vertex



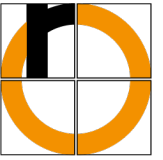
parallel edges:

Edges are incident on the same vertex

Simple (*schlichter*) graph: has neither loops nor parallel edges

A (non-directed) graph G consists of a

- set of **vertices** (or **nodes**, *Knoten*) V
- set of **edges** (*Kanten*) E
- **incidence mapping** I that maps edges to vertices
- Adjacency (*Adjazenz*)
 - the two nodes a, b of an edge e are called **adjacent**
- Incidence (*Inzidenz*)
 - the edge e that connects the nodes a, b , is **incident** on a and b
- If V is countably infinite, then G is called an **infinite graph**

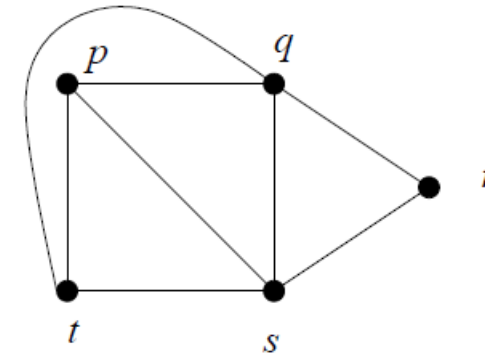
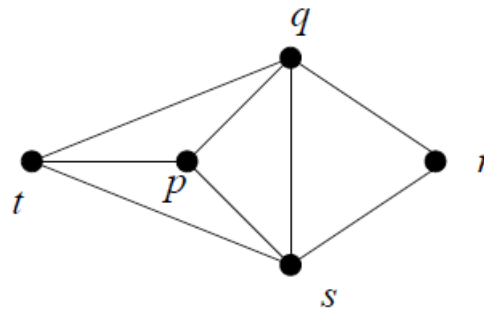
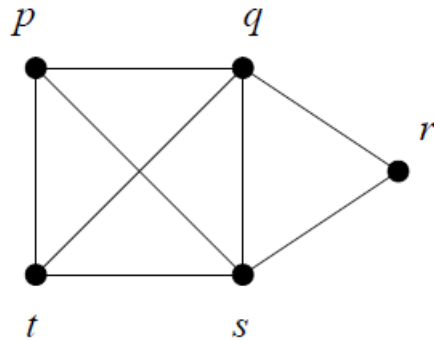


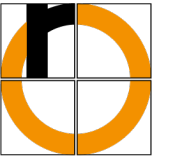
- Graphs are visualized by drawing vertices and edges
- There are many different diagrams for the same graph – don't confuse the graph with its drawing

$$V = \{p, q, r, s, t\}$$

$$E = \{\{p, q\}, \{p, s\}, \{p, t\}, \{q, r\}, \{q, s\}, \{q, t\}, \{r, s\}, \{s, t\}\}$$

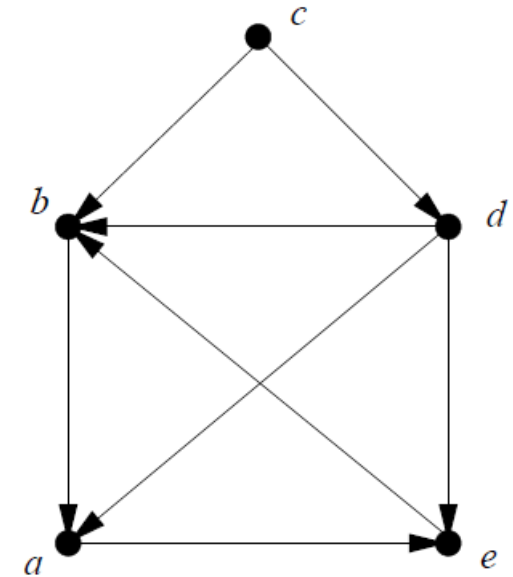
Simplified notation for simple graphs

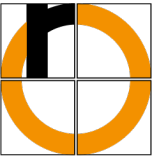




A directed (*gerichteter*) graph G consists of a

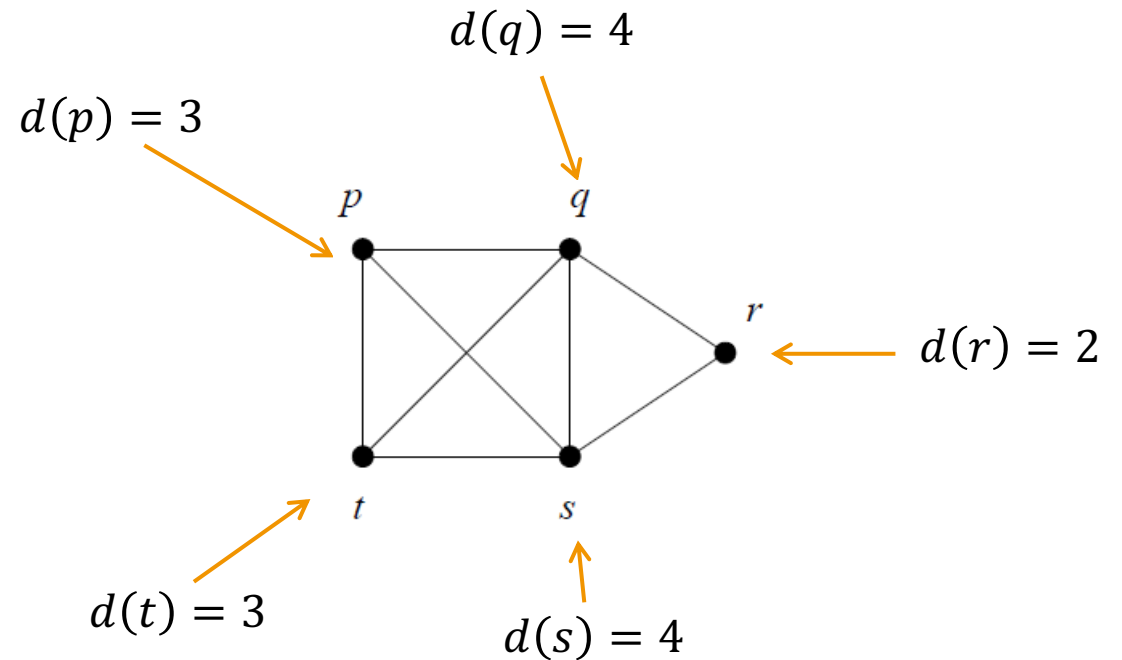
- set of vertices V
- set of **directed edges** E
 - consisting of ordered pairs of vertices $(u, v) \in V \times V$
 - u is called **start node**
 - v is called **end node**

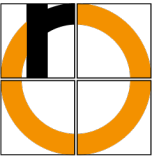




- **degree** (*Grad*) of vertex x_i :
 $d(x_i)$ = number of incident edges
- Degree sum formula
for a graph with n vertices and k edges:

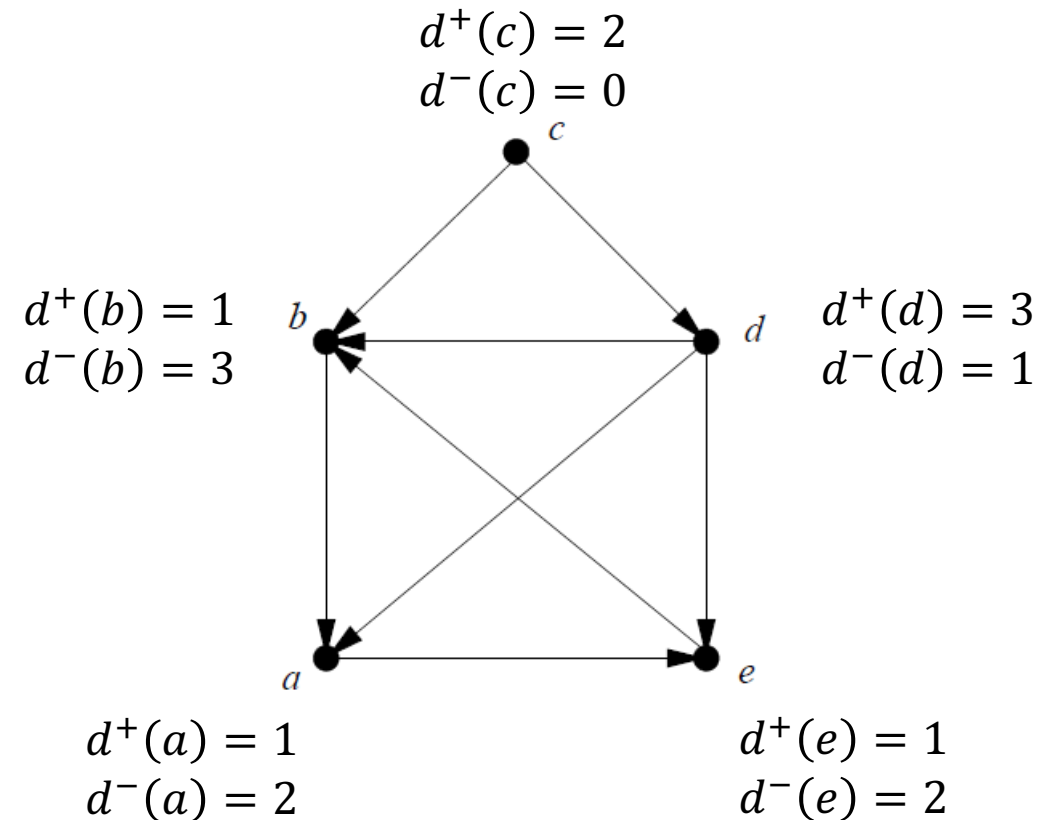
$$\sum_{i=1}^n d(x_i) = 2k$$



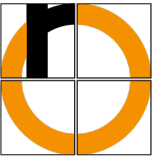


- **outdegree** (*Ausgangsgrad*)
 $d^+(x_i)$ = number of edges starting at x_i
- **indegree** (*Eingangsgrad*)
 $d^-(x_i)$ = number of edges ending at x_i
- Degree sum formula
for a graph with n vertices and k edges:

$$\sum_{i=1}^n d^+(x_i) = \sum_{i=1}^n d^-(x_i) = k$$

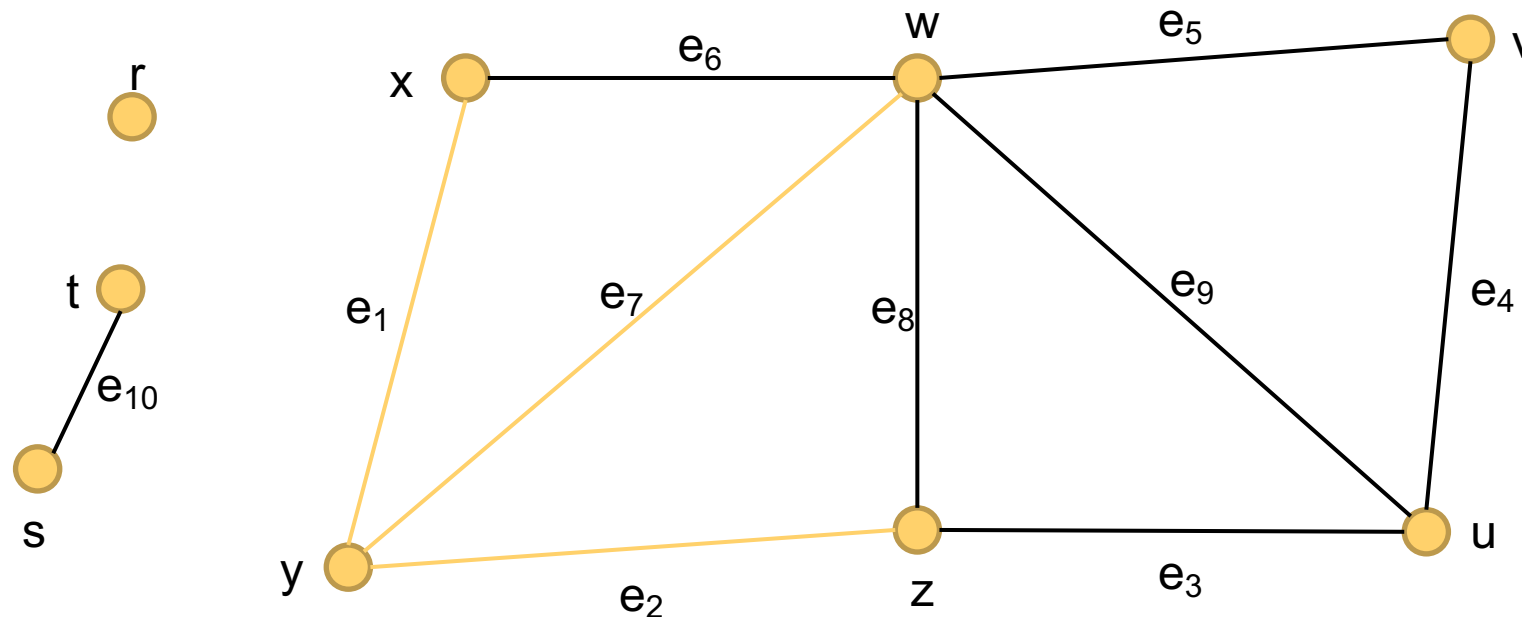


- a graph is called **complete** (*vollständig*) if there is an edge from each node to each other
- a complete (undirected) graph with n nodes has $\binom{n}{2}$ edges



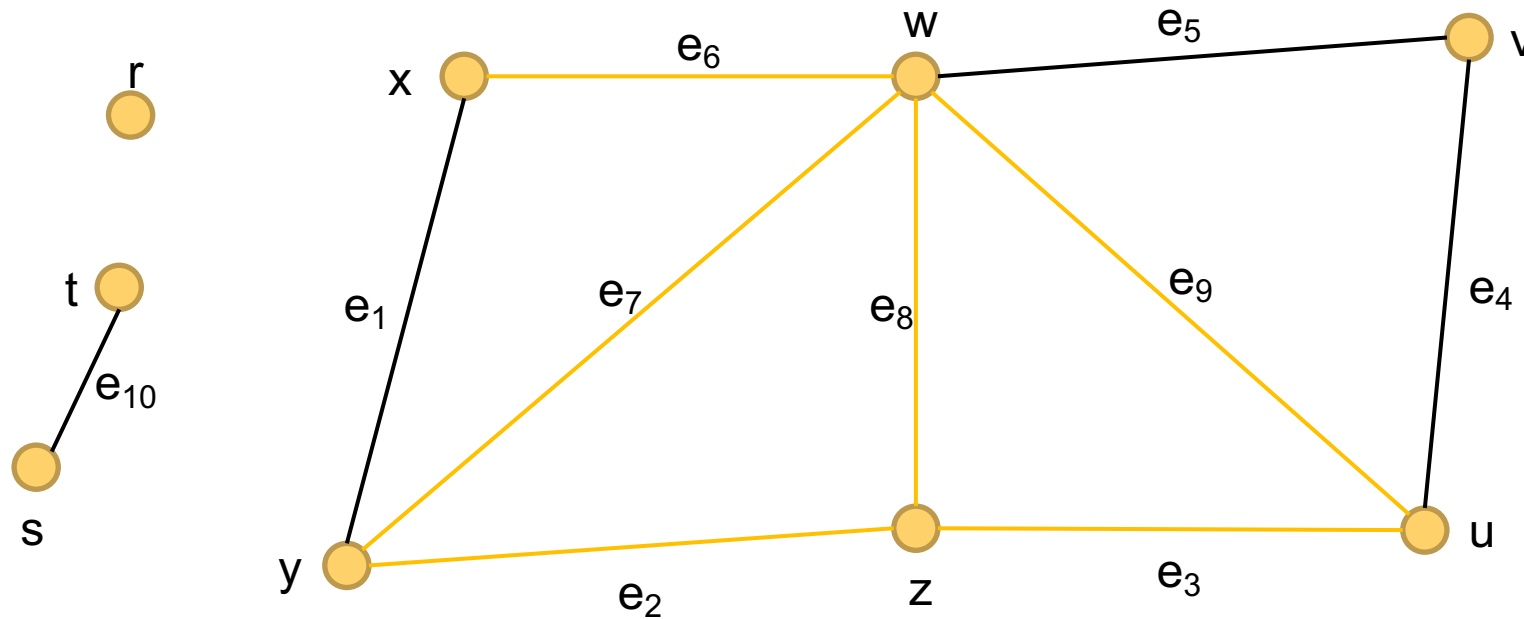
A sequence of adjacent edges from vertex v_0 to v_n (v_0, v_1), (v_1, v_2), ..., (v_{n-1}, v_n) is called a **walk** (*Kantenfolge, Kantenzug*) of length n

- Edges and vertices may be repeated
- **closed** walk: $v_0 = v_n$

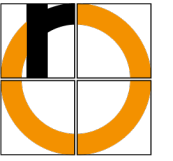


Walk from x to z (but not a trail or path): e_1, e_7, e_7, e_2 (x, y, w, y, z)

Trail (*Weg*): A walk, where all *edges* are pairwise disjoint

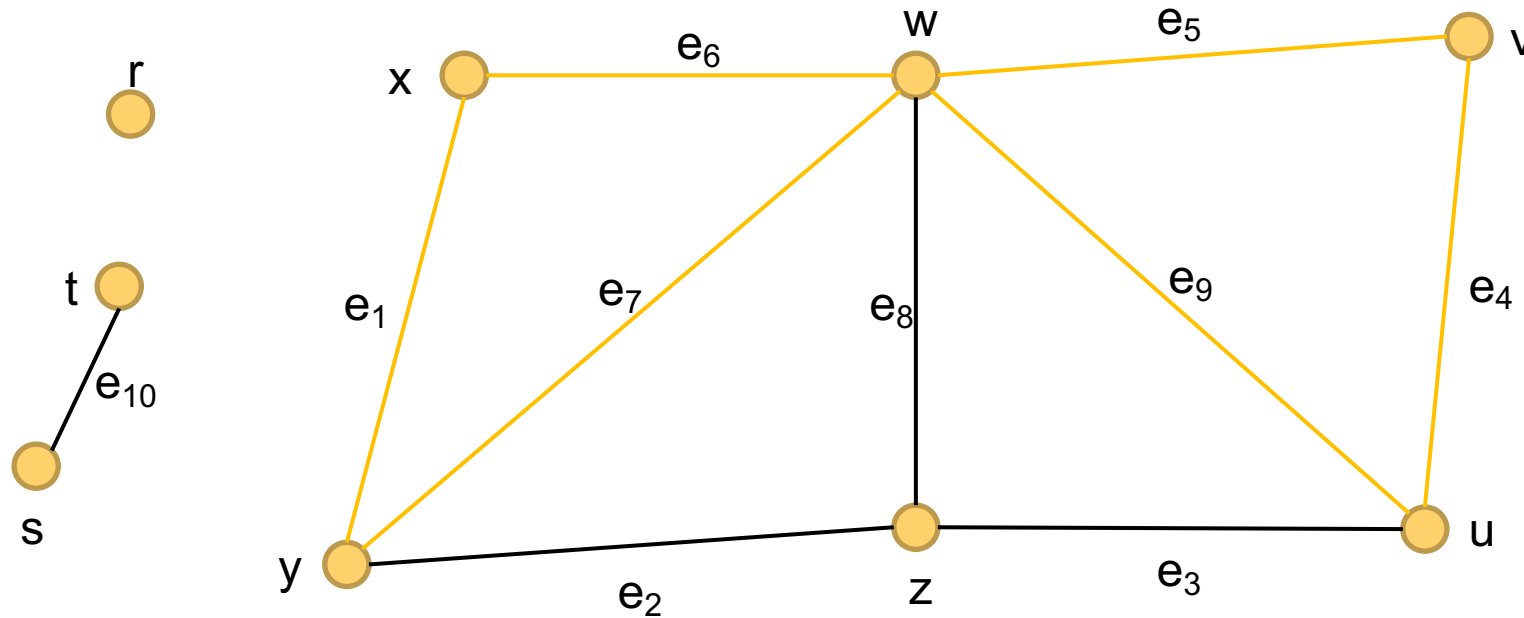


Trail from x to z (but not a path): $e_6, e_8, e_3, e_9, e_7, e_2$ (x, w, z, u, w, y, z)

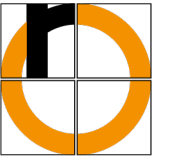


Trail (Weg): A walk, where all **edges** are pairwise disjoint

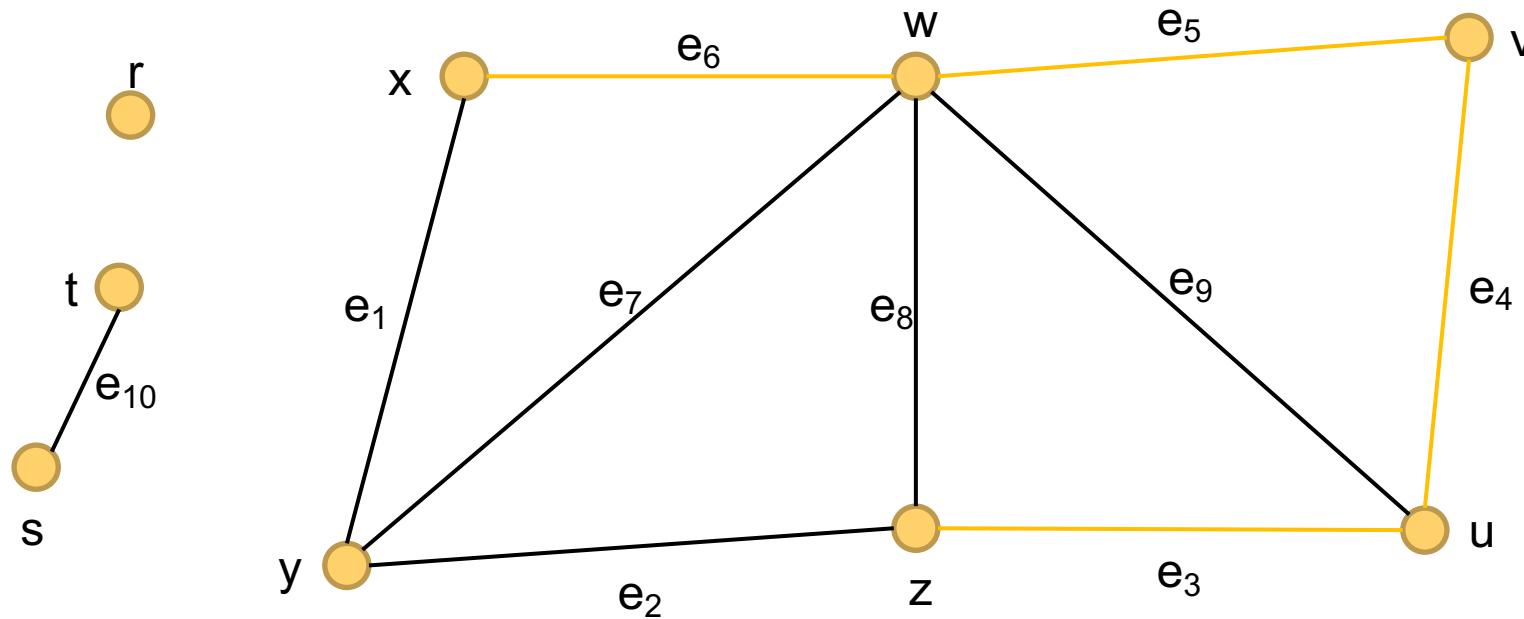
- **Closed trail (Kreis):** $v_0 = v_n$



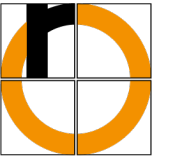
Closed trail (but not a cycle): $e_6, e_5, e_4, e_9, e_7, e_1$ (x, w, v, u, w, y, x)



Path (*Pfad*): A walk, where all *vertices* are pairwise disjoint

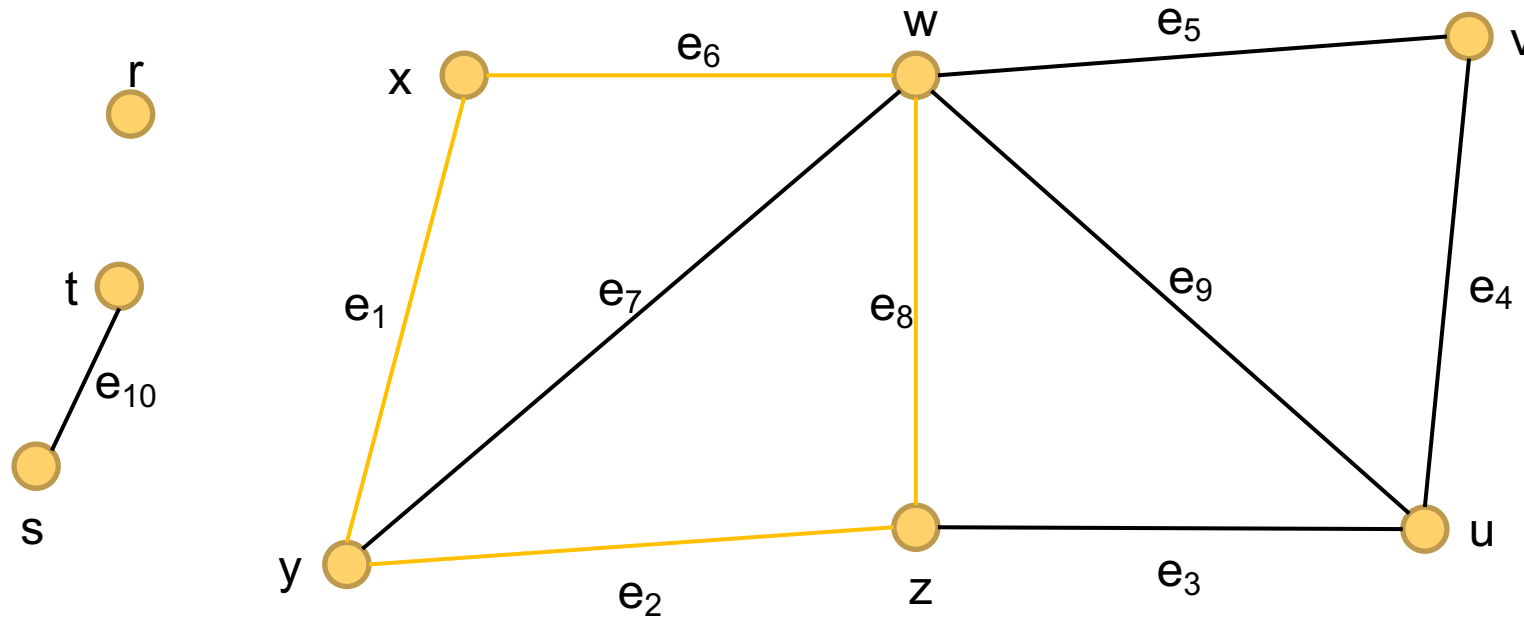


Path from x to z: e_6, e_5, e_4, e_3 (x, w, v, u, z)



Path (Pfad): A walk, where all *vertices* are pairwise disjoint

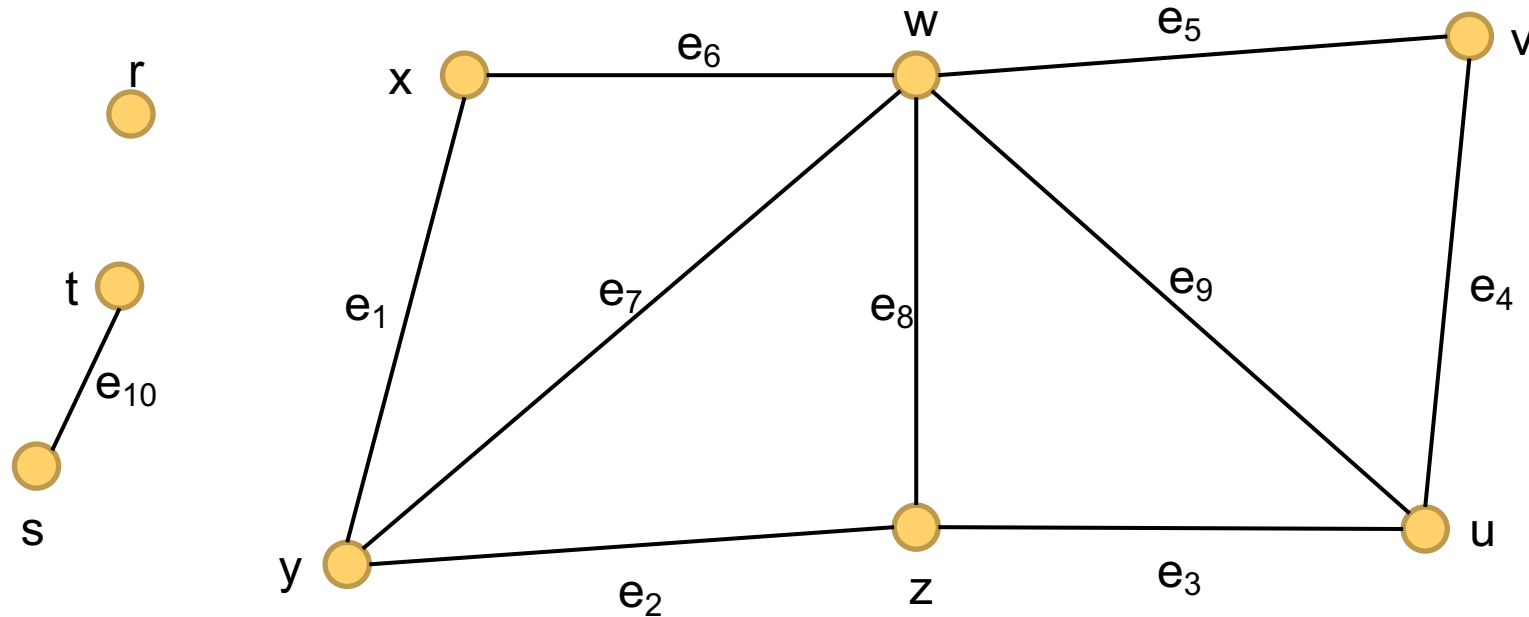
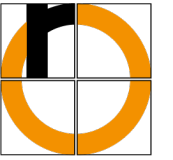
- **Cycle (Zyklus):** closed path $v_0 = v_n$
(Start-/end nodes are exempt from the rule that all vertices must be pairwise disjoint)



Cycle: e_6, e_8, e_2, e_1 (x, w, z, y, x)

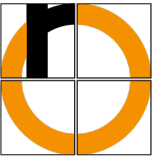
- A sequence of adjacent edges from vertex v_0 to v_n $(v_0, v_1), (v_1, v_2), \dots, (v_{n-1}, v_n)$ is called a **walk** (*Kantenfolge, Kantenzug*) of length n
 - edges and vertices may be repeated
 - **closed** walk: $v_0 = v_n$
- **Trail** (*Weg*): A walk, where all **edges** are pairwise disjoint
 - **Closed trail** (*Kreis*): $v_0 = v_n$
- **Path** (*Pfad*): A walk, where all **vertices** are pairwise disjoint
 - **Cycle** (*Zyklus*): closed path $v_0 = v_n$
(Start-/end nodes are exempt from the rule that all vertices must be pairwise disjoint)

Note: These terms are not used consistently in the literature

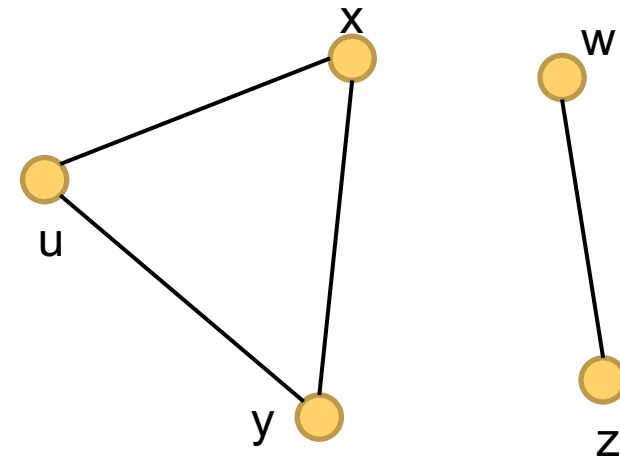
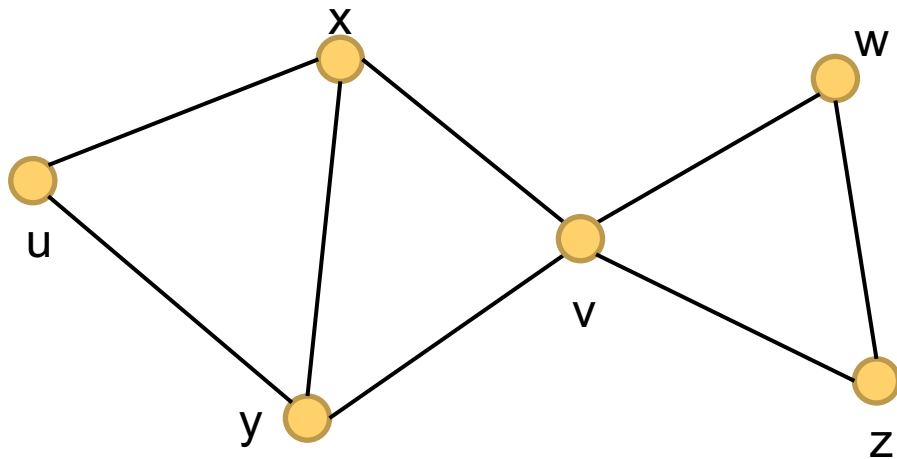


- r is an **isolated** vertex
- s and t are connected
- s and y are not connected
- This is **one** graph. The **graph** is **disconnected**
- it consists of three **connected components**
 - {r}
 - {s, t}
 - {x, y, z, u, v, w}

- Two **nodes** v, w are called **connected** (*verbunden*) if there is a path from v to w
- A graph G is said to be **connected** (*zusammenhängend*) if and only if all pairs of vertices of G are connected
 - each connected graph with n vertices has at least $n - 1$ edges
- A **connected component** (*Zusammenhangskomponente*) of G is a connected subgraph $G(U)$ induced by a set of vertices $U \subseteq V$ that has the maximum number of vertices possible

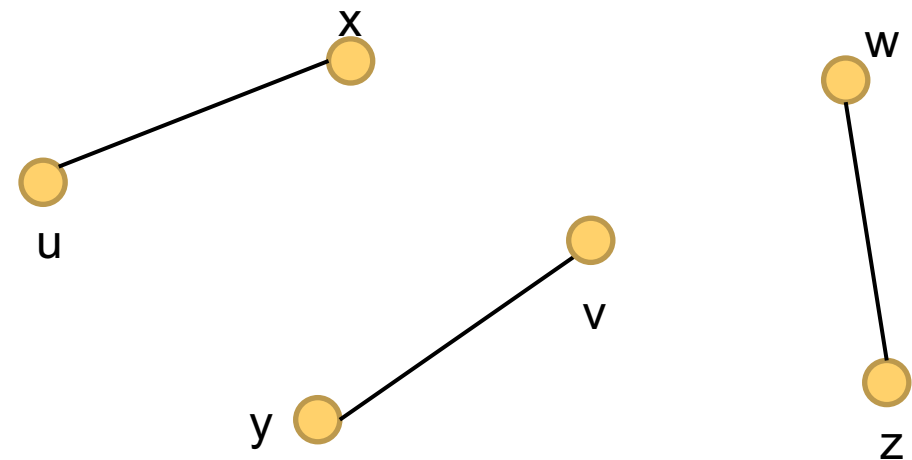
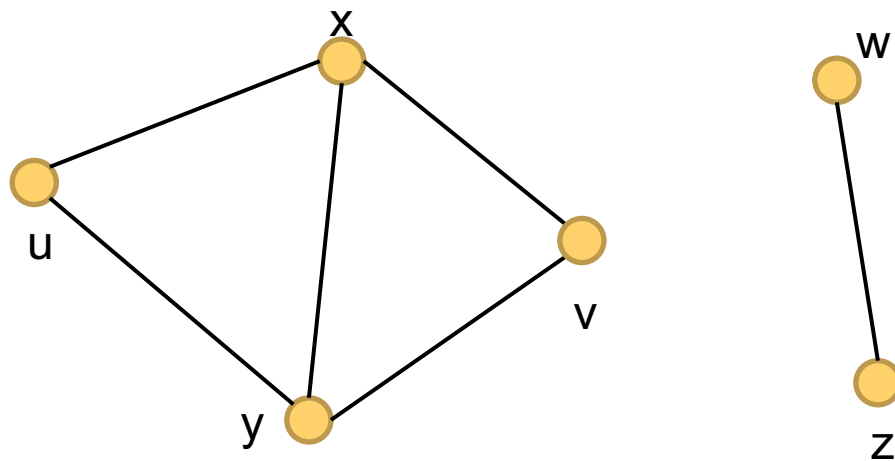
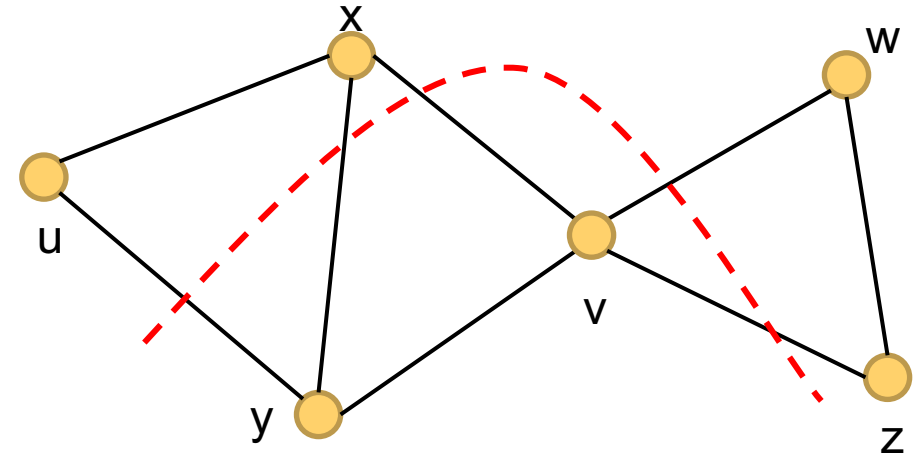
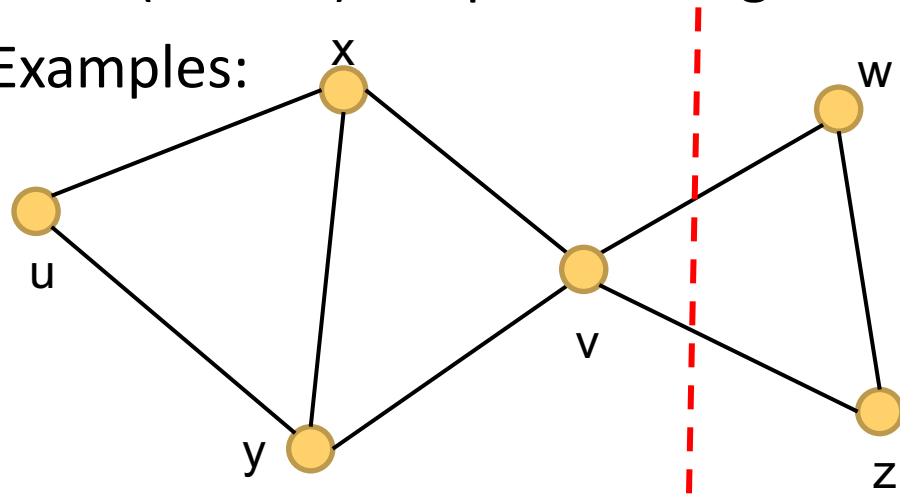


- A **vertex separator** (*Trenner*) is a **subset of vertices** of a graph that separates the graph into distinct connected components if removed (together with the incidental edges).
- Special case: A single vertex is called **separating** (*trennend*) if, after removing this vertex (and the incidental edges), the residual graph has more components than before.
- Examples
 - there are no separating nodes in the graph on the previous slide
 - in the following graph, only v is a separating vertex:



- A **cut** (*Schnitt*) is a partitioning of a graph induced by removing a **set of edges (cut-set)**

- Examples:

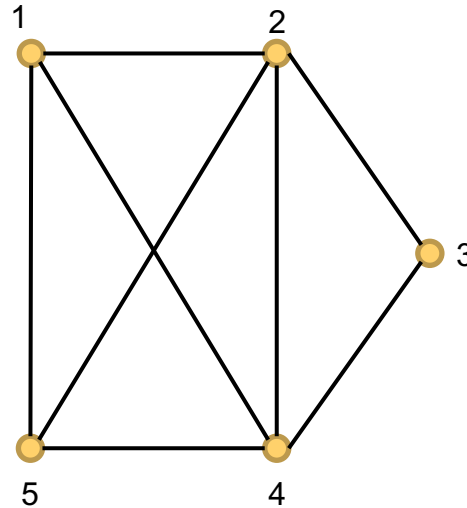
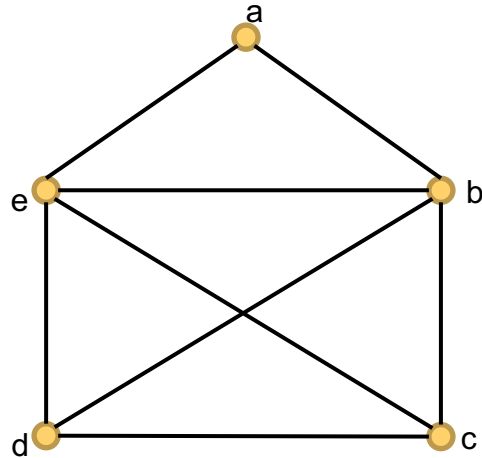
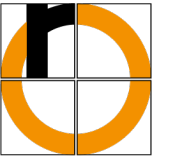


- Two graphs G_1 and G_2 are said to be **isomorphic** (*isomorph*) if and only if there exists a bijective mapping h of vertex set V_1 to V_2 such that

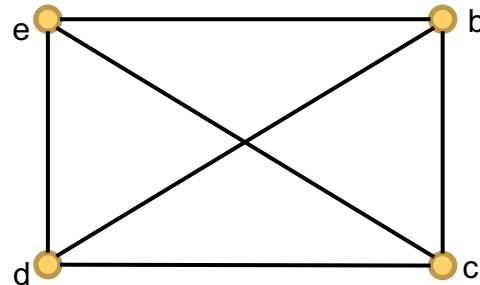
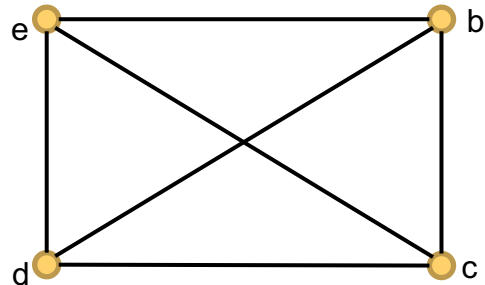
$$\forall v, w \in V_1: \{v, w\} \in E_1 \Leftrightarrow \{h(v), h(w)\} \in E_2$$

- i.e., if (v, w) is an edge of G_1 then $(h(v), h(w))$ is an edge of G_2
 - G_2 emerges from G_1 by renaming the vertices
 - isomorphic graphs have the same properties
- h is called **isomorphism** and denoted as $G_1 \simeq G_2$

Graph Isomorphism – Example

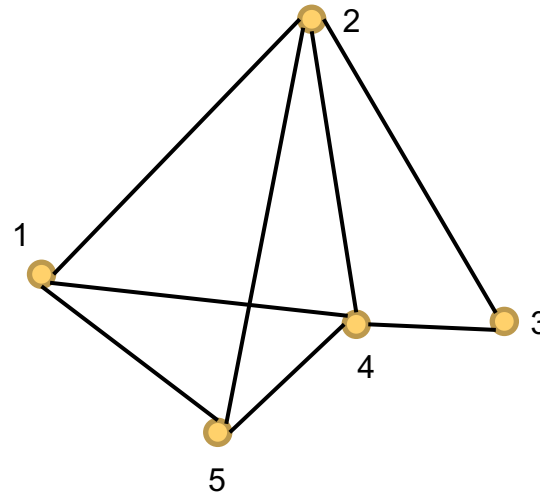
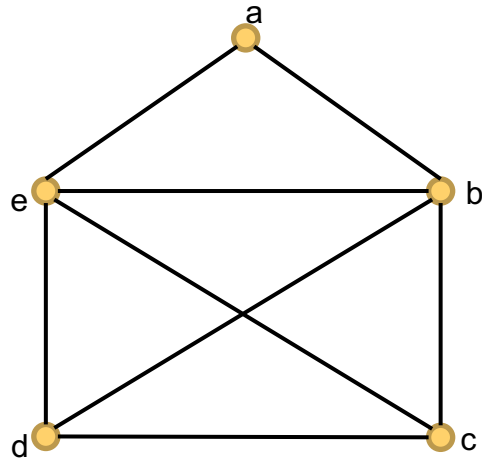
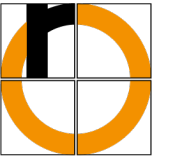


isomorphic: nodes renamed (and drawn rotated)

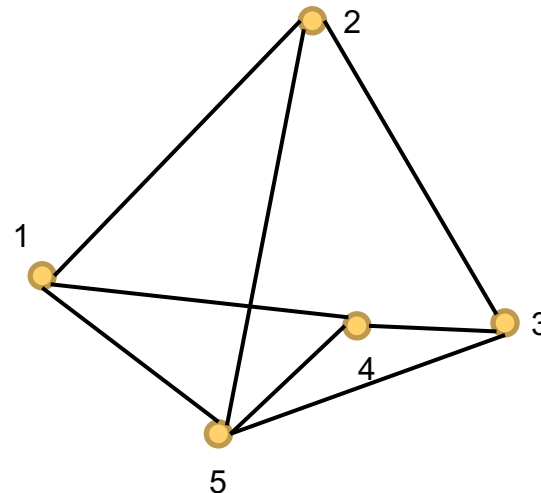
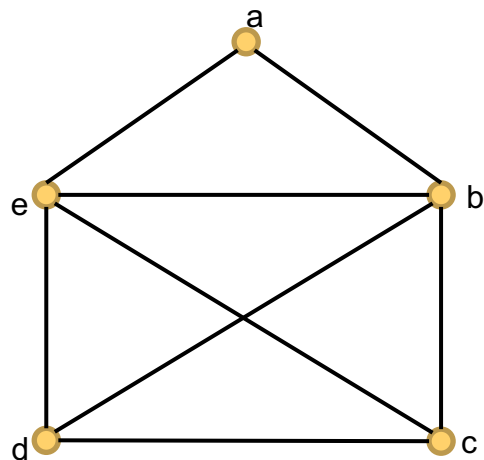


not isomorphic:
different number of vertices and edges

Graph Isomorphism – Example

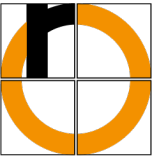


isomorphic: vertices renamed and moved

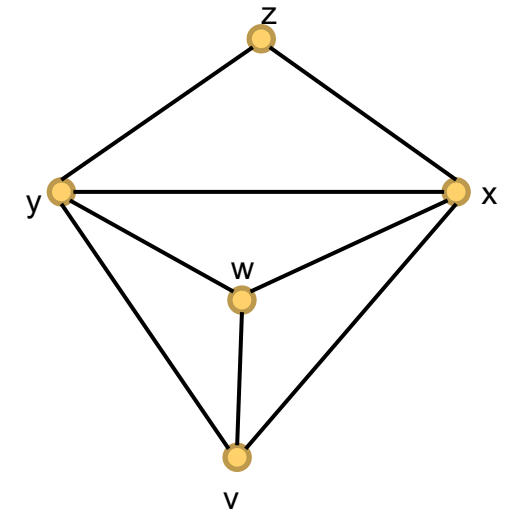
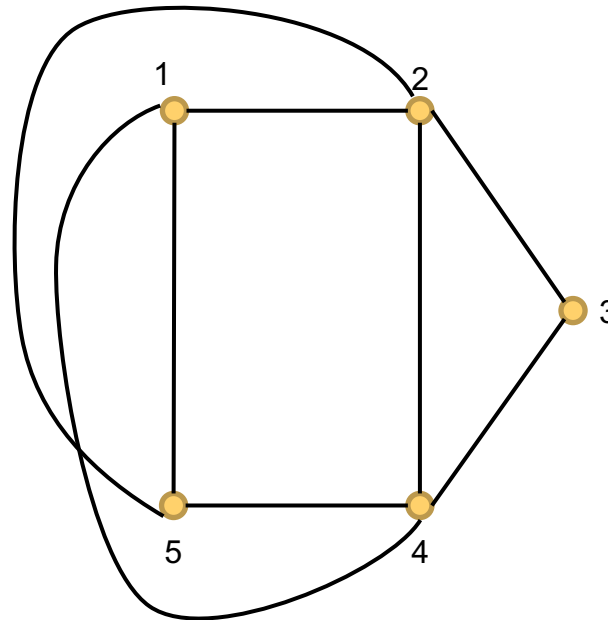
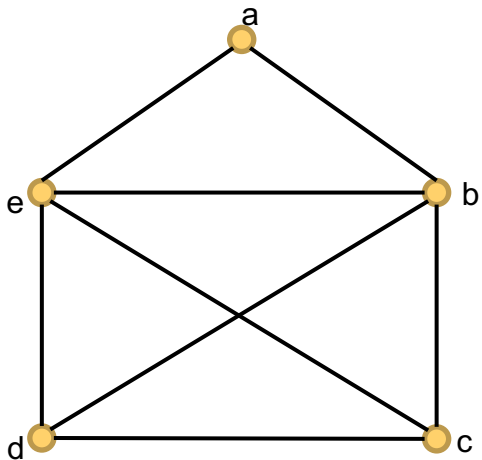


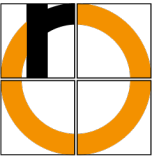
not isomorphic:
same number of vertices & edges,
but connected differently

Graph Isomorphism – Exercise

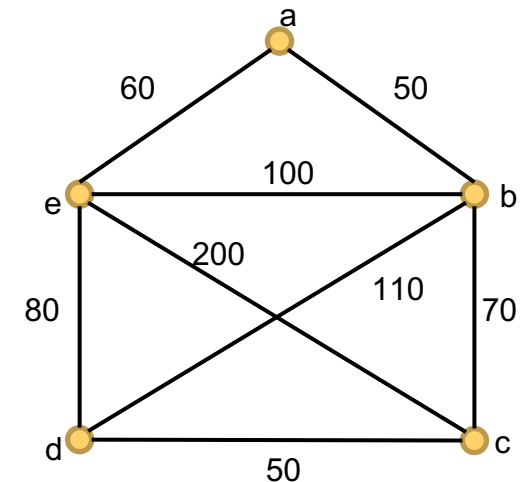


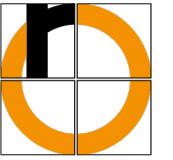
Which of the following graphs are isomorphic?
In case of isomorphism, give a bijective mapping of the vertices!



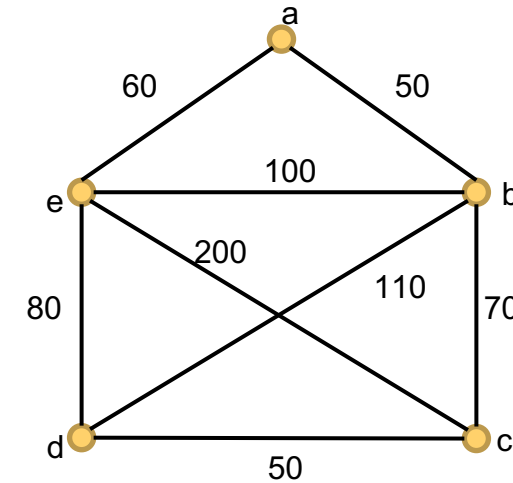


- Assign values to the edges of a graph: this is called a **weighted** (gewichteter) graph
- Examples:
 - Distances/Lengths
 - Time
 - Costs
 - Probabilities
- in some cases, negative weights can be useful
 - however, these lead to problems with distance calculations
 - therefore, it is assumed here that weights are not negative
- unweighted graph: special case, all weights equal 1





- **Length** of a walk: Sum of all edge weights
- **Distance** $\delta(v, w)$ of two vertices v, w :
 - Minimum of all walks from v to w
 - If there is no walk: $\delta(v, w) = \infty$



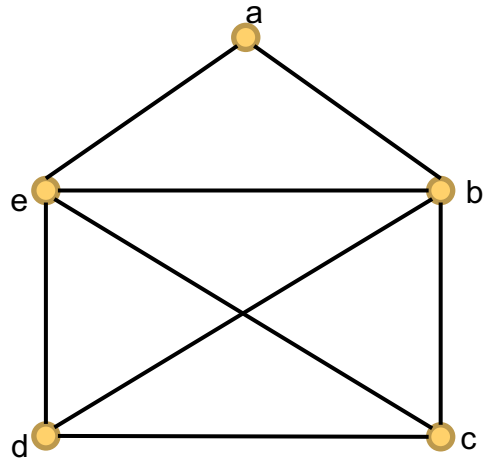
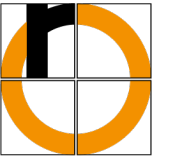
Distance between a and d: 140

- Representation of a graph in matrix form
- Graph with n vertices results in an $n \times n$ matrix A
- The elements a_{ij} of A for a given labeling of the vertices are

$$a_{ij} = \begin{cases} 1 & \text{if } (x_i, x_j) \text{ is an edge of the graph} \\ 0 & \text{otherwise} \end{cases}$$

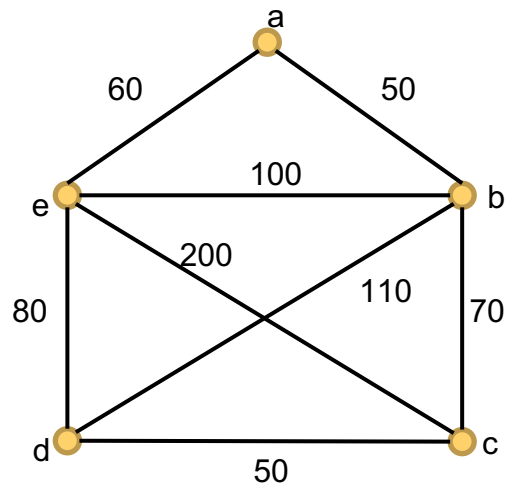
- A is called **adjacency matrix** (*Adjazenzmatrix*) of the graph
 - symmetric for non-directional graphs
 - in general asymmetric for directed graphs
 - weighted graphs: Use edge weights instead of 0 and 1

Adjacency Matrix – Example



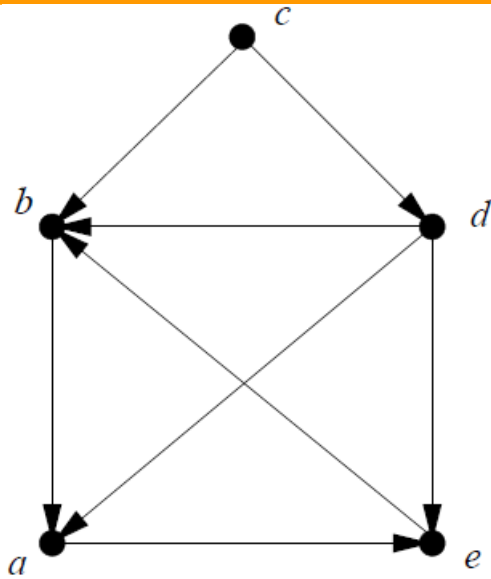
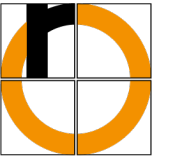
$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix} \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix}$$

order of vertices is arbitrary!

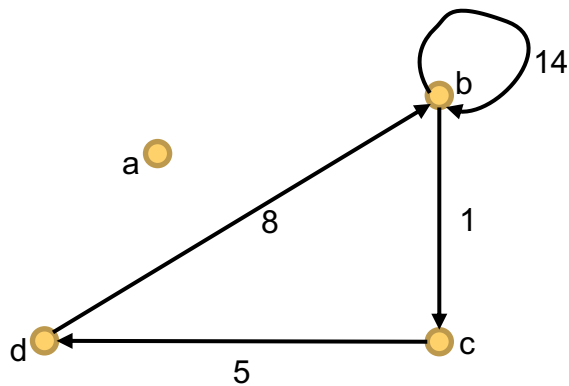


$$A = \begin{pmatrix} 0 & 50 & 0 & 0 & 60 \\ 50 & 0 & 70 & 110 & 100 \\ 0 & 70 & 0 & 50 & 200 \\ 0 & 110 & 50 & 0 & 80 \\ 60 & 100 & 200 & 80 & 0 \end{pmatrix}$$

Adjacency Matrix – Example/Exercise



$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

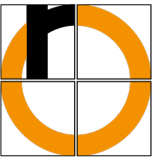


$$A = ?$$

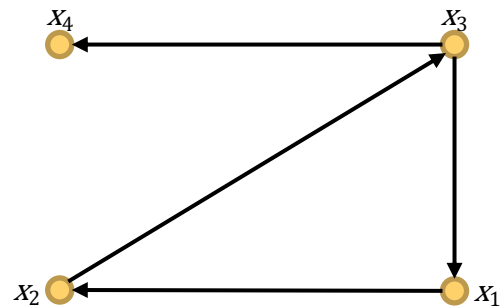
- Powers A^r of the adjacency matrix A give us information about **existence** and **number** of **walks** in **directed** graphs
- Number of different walks of length r from x_i to x_j = element a_{ij} of matrix A^r
- Graph with n vertices is **acyclic** (*azyklisch*), if there exists an r with $1 \leq r < n$ such that:
 $A^r \neq 0$, but $A^s = 0 \forall s > r$

This is said to be a **Directed Acyclic Graph** (DAG)

Powers of the Adjacency Matrix – Example



Graph with n vertices is **acyclic** (*azyklisch*), if there exists an r with $1 \leq r < n$ such that $A^r \neq 0$, but $A^s = 0 \forall s > r$



$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A^4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

all powers A, A^2, A^3, A^4 are unequal $0 \Leftrightarrow$ graph has cycles

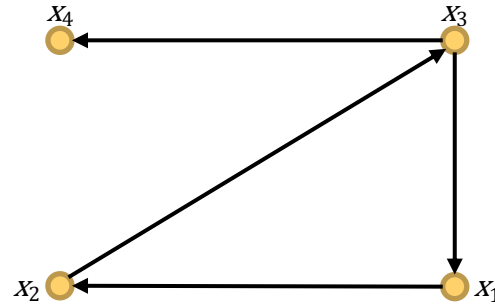
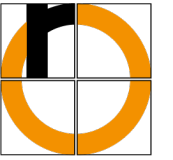
- The **path matrix** (*Wegematrix*) W indicates whether a path from x_i to x_j exists:

$$w_{ij} = \begin{cases} 1 & \text{if there exists a path from } x_i \text{ to } x_j \\ 0 & \text{otherwise} \end{cases}$$

- W can be obtained by adding up all relevant powers of the adjacency matrix

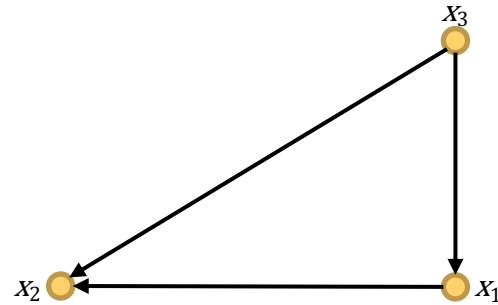
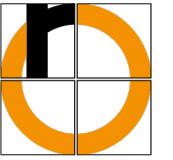
$$A + A^2 + A^3 + \dots + A^n$$

and replacing all non-zero elements by 1



$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \mathbf{A}^3 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \mathbf{A}^4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \longrightarrow \mathbf{W} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$



1. Determine the adjacency matrix and its powers.
2. What can be said about the graph from these results?
3. Determine the path matrix

- Adjacency matrix
 - undirected graphs: it is sufficient to store half the matrix
 - often contains many zeros
- Adjacency list
 - Linked list of vertices
 - for each vertex: contains a linked list of its neighbors
 - more compact than the adjacency matrix

Adjacency List – Example

